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Started	Friday, 30 October 2026, 1:22 AM
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Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It allows for faster searching
- b. It increases the number of comparisons needed
- c. It has no effect on the search operation
- d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **2**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Divide and conquer
- b. Greedy method
- c. Dynamic programming
- d. Backtracking

The correct answer is:
Divide and conquer

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It repeatedly finds the minimum element and moves it to the sorted part of the list
b. It divides the list into two halves, sorts each half, and then merges them
c. It compares adjacent elements and swaps them if necessary
d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **4**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By breaking the input into smaller parts and merging them
b. By sorting data in a single pass
c. By comparing elements sequentially
d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **5**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It recursively sorts subproblems
b. It divides the list into sublists
c. It bubbles up the largest element in each pass
d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **6**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Merging
b. Rearranging
c. Sorting
d. Searching

The correct answer is: Sorting

Question **7**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Binary Search
b. Linear Search
c. Merge Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is less efficient than bubble sort
b. It sorts data using a single pass
c. It is based on the divide-and-conquer approach
d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **9**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It only works with integer arrays
b. It is less efficient than custom sorting algorithms
c. It sorts data out of the box efficiently
d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently

Question **10**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. All of the above
b. Selection
c. Duplicates
d. Searching

The correct answer is: All of the above

Question **11**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Bubble sort
b. Heap sort
c. Insertion sort
d. Quick sort

The correct answer is:
Insertion sort

Question **12**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sort()
- b. order()
- c. arrange()
- d. sorted()

The correct answer is: sort()

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Merging two datasets
- b. Arranging data in a specific order
- c. Removing duplicates from a list
- d. Finding a specific element in a list

The correct answer is: Arranging data in a specific order

Question **14**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Sorting data in a single pass
- b. Sorting data sequentially
- c. Dividing the input into parts, solving each part, and combining the solutions
- d. Rearranging data without sorting

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **15**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Searching

b. Merging

c. Sorting

d. Complexity

The correct answer is:
Complexity